

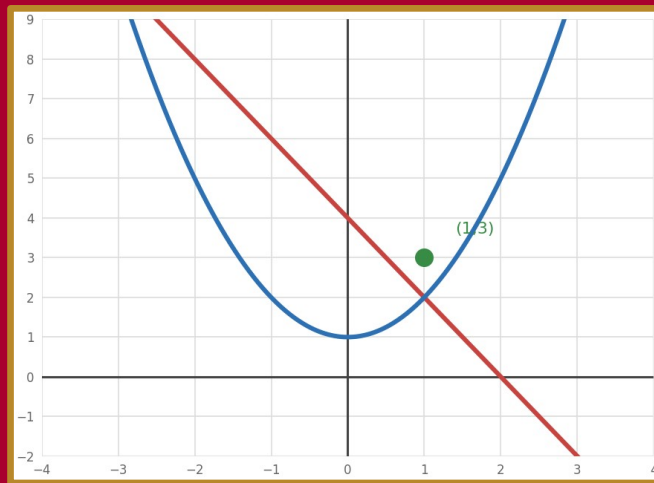


Desmos

Made Simple

The free graphing calculator, explained step by step

desmos.com/calculator



CLC Educational Institute

What is Desmos?



Free

- No cost for students
- No install needed



Runs in your browser

- Works on any laptop
- Apps for phone and tablet

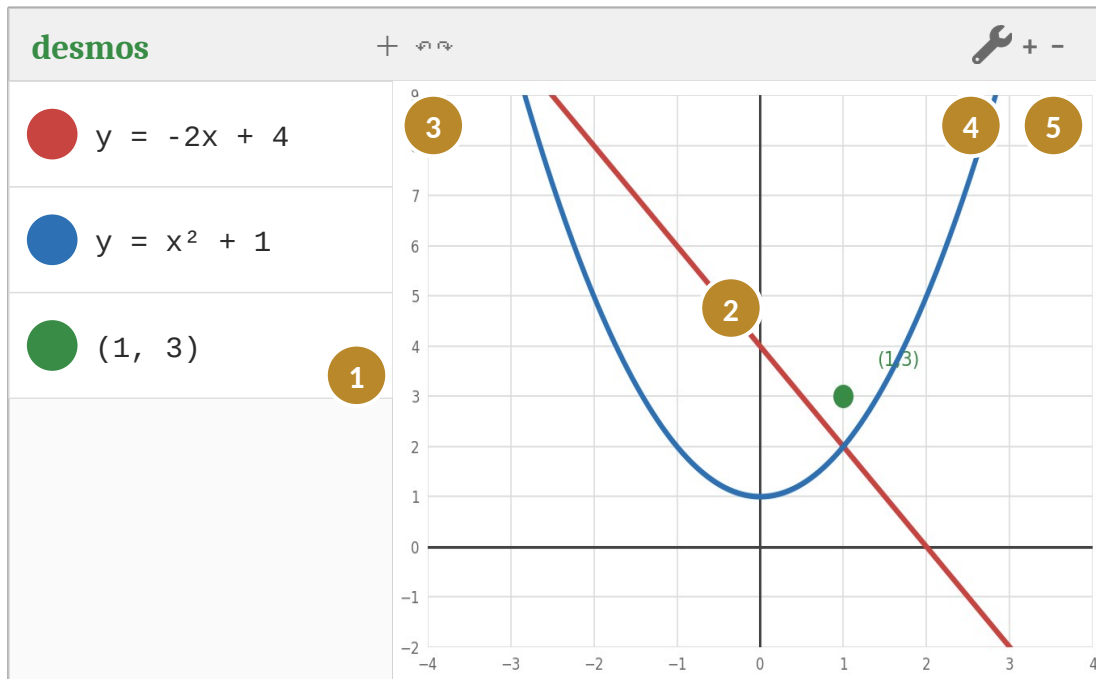


Instant graphs

- Type an equation
- The graph appears at once

Open it: desmos.com/calculator

Meet the screen



- 1 **Expression list**
Type equations, points, tables here
- 2 **Graph area**
Your graph appears here
- 3 **Add item (+)**
Adds tables, notes, folders
- 4 **Settings (wrench)**
Change axes, grid, angles
- 5 **Zoom buttons**
Zoom in, zoom out, reset

Step 1: Type it. See it.

Point

$(1, 3)$

- Draws a dot at $x=1, y=3$

Line

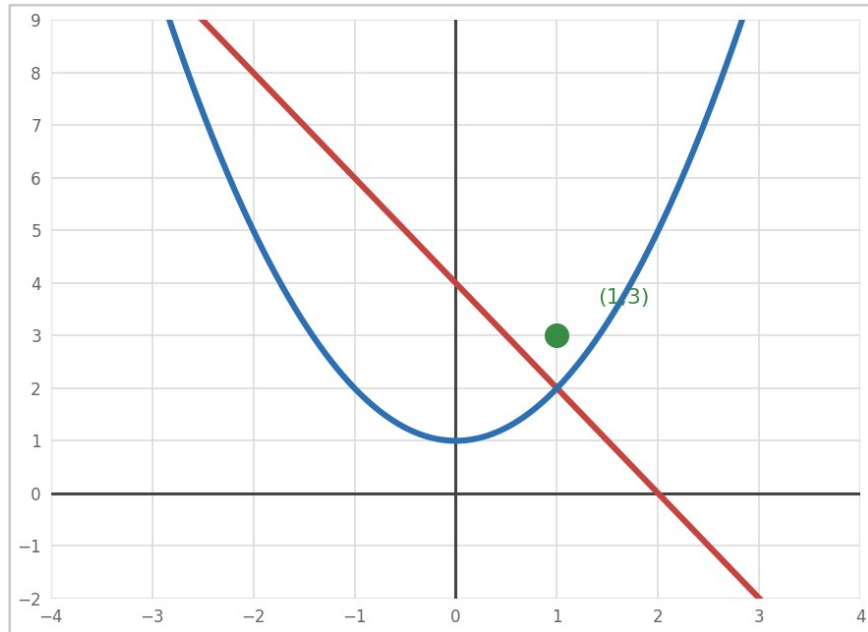
$y = -2x + 4$

- Draws a straight line

Curve

$y = x^2 + 1$

- Draws a parabola





Typing shortcuts

^

Power

x^2 gives x^2

/

Fraction

$1/2$ gives $\frac{1}{2}$

sqrt

Square root

sqrt(9) gives $\sqrt{9}$

pi

Pi

pi gives π

<=

Less or equal

<= gives $\leq$

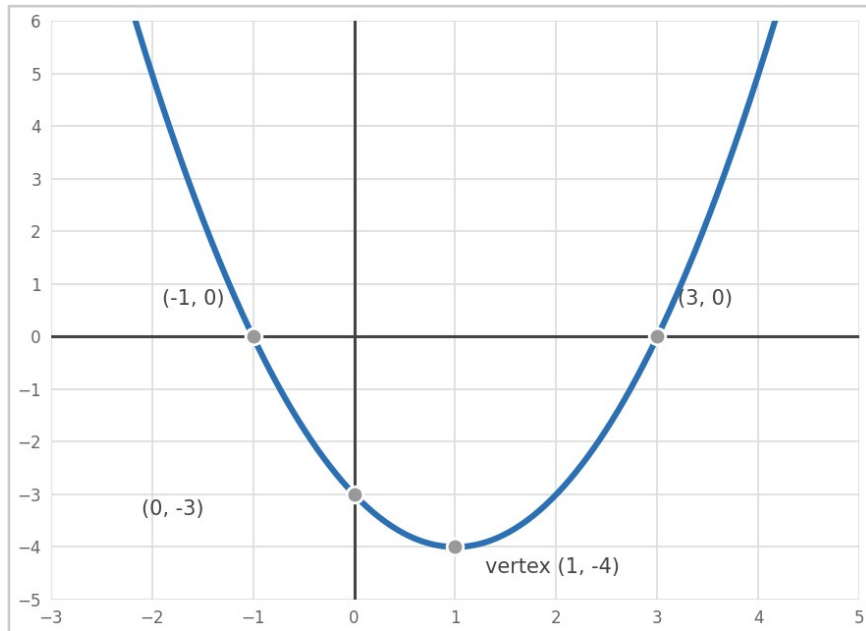
abs

Absolute value

abs(x) gives $|x|$

Stuck? Click Show Keypad for every symbol.

Find the key points



What you see

- Gray dots appear on graphs
- They mark important points

How to reveal them

- Click the curve
- Or click its expression line

What they show

- Intercepts and vertex
- Intersections of two curves

Sliders: make graphs move

Type

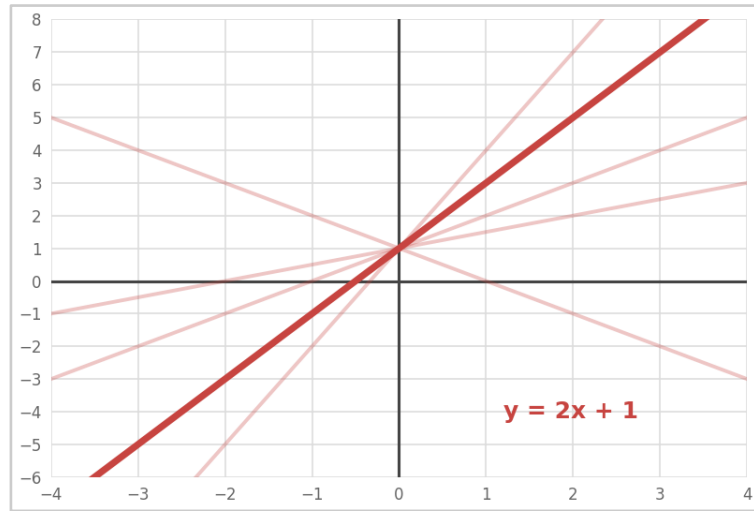
$$y=mx+b$$

Click

- Choose “add sliders” for m and b

Drag

- m changes the slope
- b moves the line up or down



m =

b =

Tables: plot your own data

1 Click + (Add item)

2 Choose Table

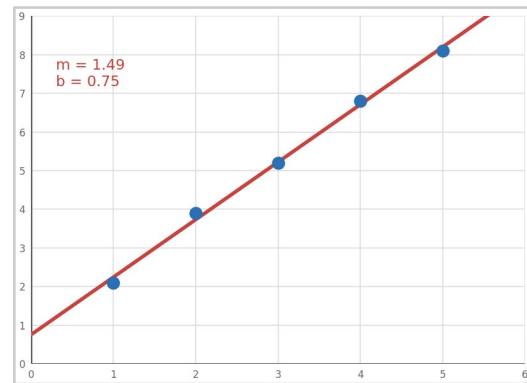
3 Type x and y values

4 Points appear on graph

x_1	y_1
1	2
2	4
3	5
4	7

Good to know

- Type values in each cell
- Paste data from a spreadsheet
- Use Zoom Fit to see all points



Best-fit line in one line

Type under your table

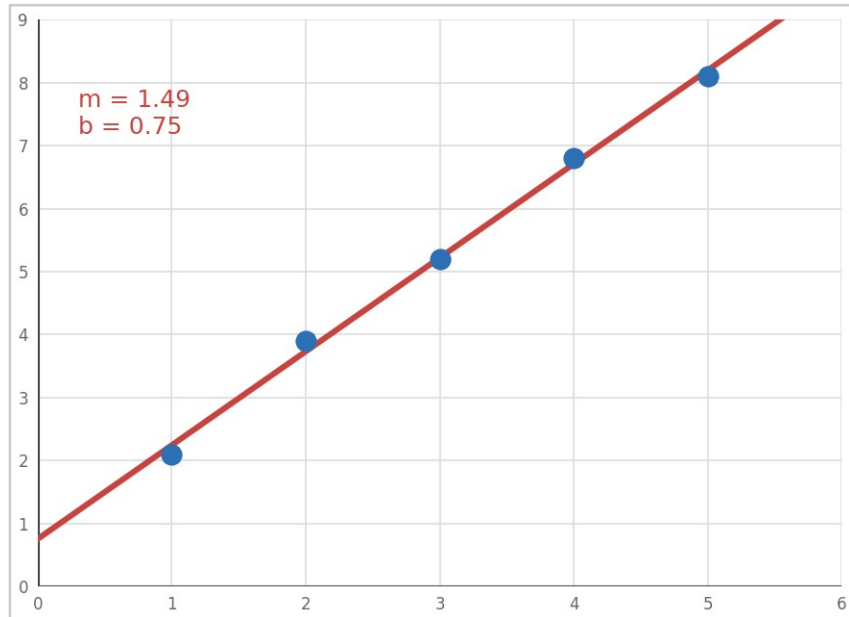
$$y_1 \sim mx_1 + b$$

The symbol

- $\sim$ is called tilde
- Find it left of the 1 key

The result

- Desmos finds the best m and b
- Fit quality is shown with r^2



Limit it. Shade it.

Limit a graph

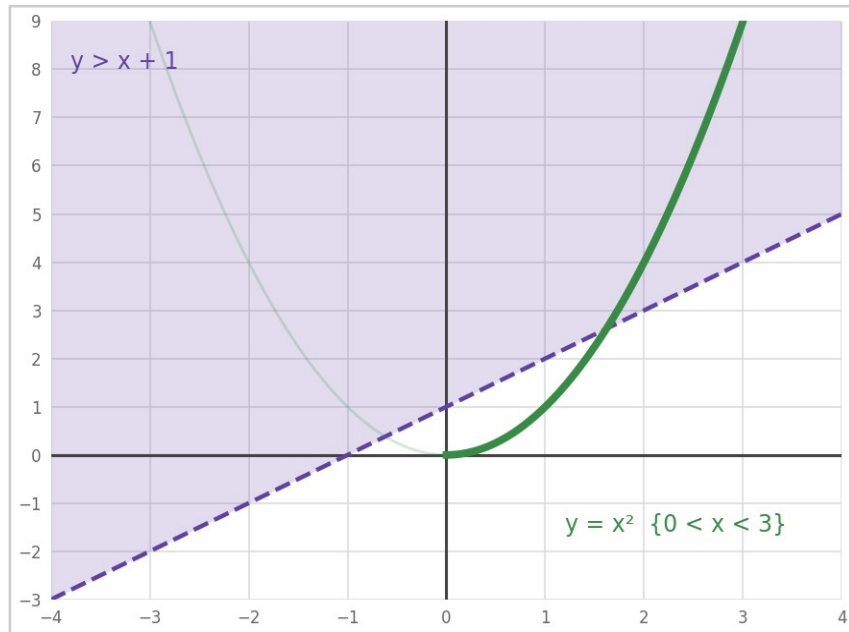
$$y = x^2 \quad \{0 < x < 3\}$$

- Curly braces set the range
- Only x from 0 to 3 shows

Shade a region

$$y > x + 1$$

- Desmos shades every point that fits



Move around the graph



Pan

- Click and drag the graph
- Touch: drag with one finger



Zoom

- Scroll wheel, or + and -
- Touch: pinch and spread



Reset view

- Home button resets
- Zoom Fit centers your data



Settings

- Wrench sets axis limits
- Also grid and angle units

Save and share



Save

- Sign in (free)
- Press Ctrl+S



Share

- Use the share button
- Send the link



Reopen

- Open Graph menu
- Pick a saved graph

Exam warning

Some tests use a restricted Desmos version. Check your exam rules first.



Try it yourself

1 Graph $y=x^2$, then $y=x^2+3$

What moved?

2 Type $y=a*x^2$ and add a slider

Drag a. What changes?

3 Graph $y=2x+1$ and $y=x^2$

Click the gray dots

4 Make a table, then $y_1 \sim mx_1 + b$

Check the best-fit line

5 Type $y < x - 2$

See the shaded region

Your Desmos cheat sheet



Graph

Type it in the list

Key points

Click the curve, see gray dots

Sliders

Use letters like m and b

Data

Add a table

Best fit

$y_1 \sim mx_1 + b$

Limits

Use { } after the equation

[desmos.com/calculator](https://www.desmos.com/calculator)

Sources: Desmos Help Center and Desmos User Guide